

VENANCE EPAFRA SWAI

Software Developer

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PROFESSIONAL SUMMARY

Software Developer with hands-on experience building full-stack web and mobile applications using Laravel, React, Flutter, and FastAPI. Completed a software development internship at NSSF and an IT support internship at DIT. Skilled in REST API design, database management, and building self-hosted AI-powered applications, with a strong interest in backend engineering and writing clean, maintainable code.

TECHNICAL SKILLS

Languages: PHP, JavaScript, Python

Frameworks & Libraries: Laravel, React, Flutter, FastAPI

Databases: MySQL, Firebase

Tools & Platforms: Git, GitHub, Postman, Linux (Ubuntu/WSL), VS Code, Vite

Concepts: REST APIs, Authentication (Bearer Tokens, Sanctum), MVC, CRUD, OOP, RBAC

PROFESSIONAL EXPERIENCE

Software Development Intern

2025

National Social Security Fund (NSSF) — Software Development Unit

- Built and deployed a full-stack corporate blog platform using Laravel and the Blade templating engine.
- Designed relational database schemas in MySQL and wrote migrations and queries to manage content distribution.
- Performed routine system maintenance and debugged environment-specific configuration issues.
- Implemented responsive, mobile-first design practices to improve site performance and accessibility.

IT Support & Network Technician Intern

2024

Dar es Salaam Institute of Technology (DIT) — Computer Department

- Maintained local area network (LAN) infrastructure, diagnosing connectivity issues and managing IP allocations.
- Performed hardware and software diagnostics, preventative maintenance, and operating system deployments across department workstations.
- Troubleshoot network hardware configurations to minimize downtime for critical school infrastructure.
- Assisted senior IT staff with enforcing data security baselines and managing user access controls.

KEY PROJECTS

University Student Support Assistant

AI Course Project

Tech Stack: FastAPI, React (Vite), Ollama (LLaMA 3.2), Python, Git

- Built a full-stack, self-hosted AI chatbot that answers university service queries by deploying a local LLM via Ollama, removing reliance on external APIs such as OpenAI while maintaining full data privacy.
- Engineered a FastAPI backend with structured endpoints (/health, /ask, /feedback, /upload), implementing CORS middleware, request validation, error handling, and automated logging of all interactions to a persistent log file.
- Developed a responsive React frontend featuring multi-session chat history, dark mode, document-based Q&A (file upload with in-context answering), and a user feedback rating system (Good/Average/Poor) persisted to the backend.
- Handled the complete pipeline from LLM setup and backend API design to frontend integration, an automated pytest test suite, and technical documentation.

Student Attendance System (SAS)

Final Year Capstone Project

Tech Stack: Flutter, Firebase, NFC Technology, Git

- Co-developed a mobile application that automates academic attendance tracking for 100+ students by using smartphone NFC capabilities to read student ID tags.
- Integrated Firebase for real-time data synchronization, secure user authentication, and instant cloud database updates.
- Collaborated within an Agile project team using Git/GitHub for version control, managing pull requests and resolving branch conflicts.

Inventory Lite

Personal Project

Tech Stack: PHP (Laravel), MySQL, Eloquent ORM, Git

- Built a lightweight inventory management system covering 5 core entities (products, categories, suppliers, stock levels, and transactions) using clean MVC architecture.
- Managed environment configuration variables (.env) and automated relational database seeding for testing.
- Developed CRUD routes and business logic to track asset stock levels and transactions accurately.

End-to-End API Security Testing

Practical Lab

Tech Stack: Postman, REST APIs, JSON — Job Board Platform

- Designed a test suite covering 20+ API endpoints to map and validate backend routes for an online employment marketplace.
- Tested role-based access control (RBAC), verifying route protection between employers and job seekers using Bearer Token authentication.

EDUCATION

Bachelor of Science in Computer Science

2023 – Expected 2026

University of Dar es Salaam

Certificate of Introductory Military Course

2023

Ministry of Defense and National Service, Tanzania

Ordinary & Advanced Certificate of Secondary Education (CSEE / ACSEE)

2017 – 2023

Marian Boys High School

CERTIFICATIONS & PRACTICAL TRAINING

- Software Development Internship — NSSF (2025)
- IT Support Internship — DIT (2024)

LANGUAGES

English (Fluent) — Swahili (Native)

REFERENCES

Available upon request.